Credit Risk Intelligence: Early Warning Signals for Loan Defaults

A Behavioral and Financial Risk Analysis

Project By: Faraz Shaikh | Domain: Banking, Financial Services & Insurance (BFSI)

Tools Used: SQL | Python | Tableau

Business Problem

NovaBank is a digital lending company providing short-term loans ranging from ₹5,000 to ₹200,000. Over the past 12 months, the loan default rate increased from 4% to 16%, leading to significant financial losses.

This rising default rate created risk to profitability and long-term financial stability. The organization required data-driven insights to identify key risk factors and reduce loan defaults.

Key Impact

Default rate increased from 4% to 16% in 12 months, threatening financial stability and profitability.

Project Objective

Identify Risk Patterns

Analyze loan, customer, and repayment data to identify patterns leading to loan defaults.

Early Warning Signals

Identify early warning signals that indicate high-risk loans before they default.

Reduce Defaults

Provide actionable recommendations to reduce default rates and financial losses.

KPI Framework

1

Primary KPI

Default Rate (%) — Percentage of loans that defaulted versus total loans issued

2

Loan Characteristics

Loan Tenure | Loan Purpose | Interest Rate

3

Customer Profile

Income Level | Employment Type | Credit Score

4

Repayment Behavior

Missed Payments | Late Payment Ratio | Debt-to-Income Ratio

Dataset Overview

The dataset consists of three relational tables containing comprehensive loan and customer information.

01

Customer Table

Contains customer demographics and financial information

02

Loan Table

Stores loan details such as amount, tenure, interest rates and loan status

03

Payment Table

Records payment transactions and repayment behavior

Total Records

- 10,000 Customers
- 12,000 Loans
- 160,000+ Payment Records

Relationship Structure

One Customer → Multiple Loans

One Loan → Multiple Payments

Key Behavioral Finding

Repayment behavior is the strongest predictor of loan default. Loans with repeated missed payments show extremely high risk of default.

1

Clear Pattern

A clear risk escalation pattern is observed as missed payments increase.

2

Early Detection

This creates an opportunity to detect risky loans early and take preventive action.

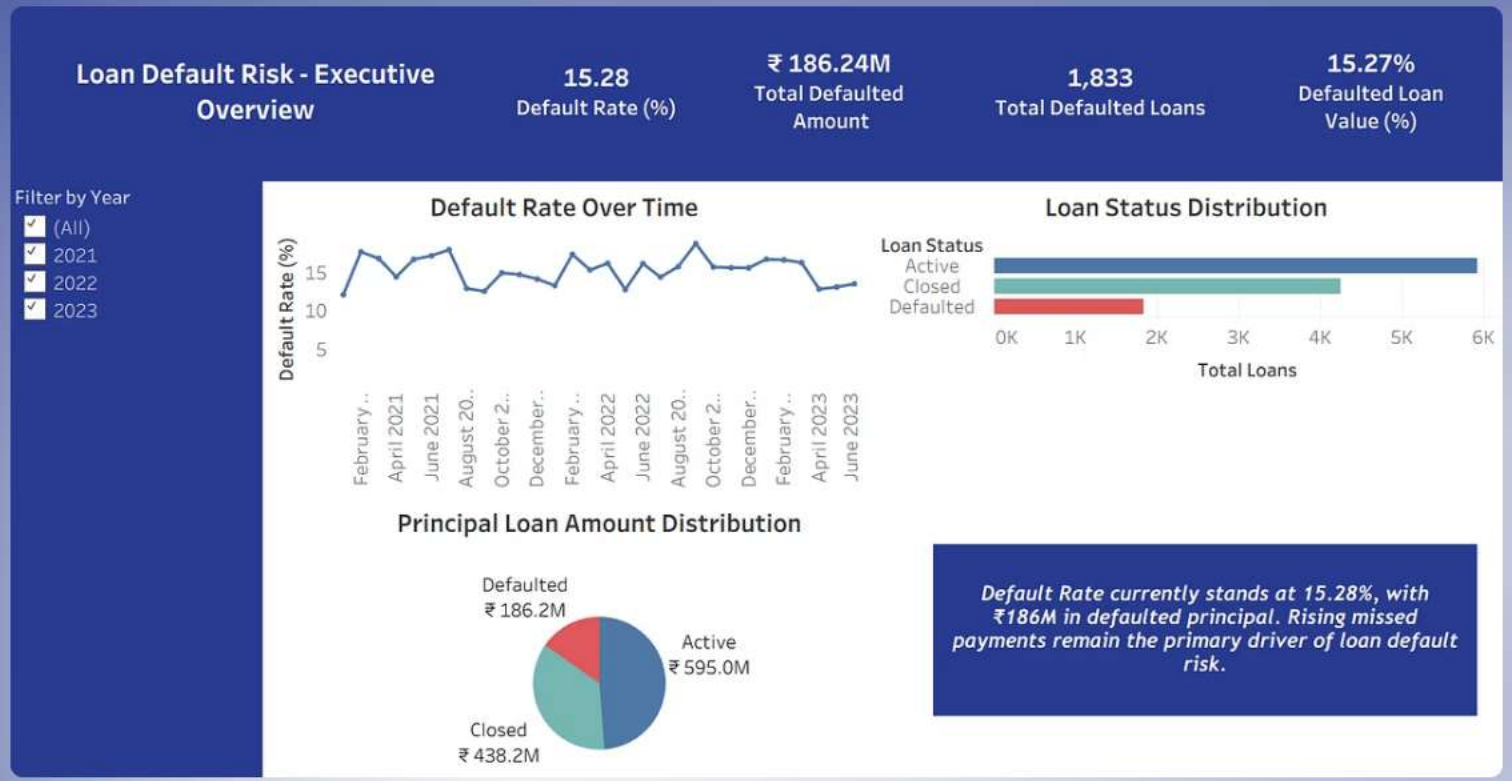
Dashboard 1: Executive Overview

Purpose

Provide a high-level summary of loan portfolio performance.

Key Metrics

- Default Rate (%)
- Total Defaulted Loans
- Total Defaulted Amount
- Percentage of Loan Value Lost



Dashboard 2: Risk Segmentation

Purpose

Understand how loan and customer characteristics impact default risk.

Key Analysis Areas

- Loan Tenure
- Credit Score
- Income Group
- Employment Type

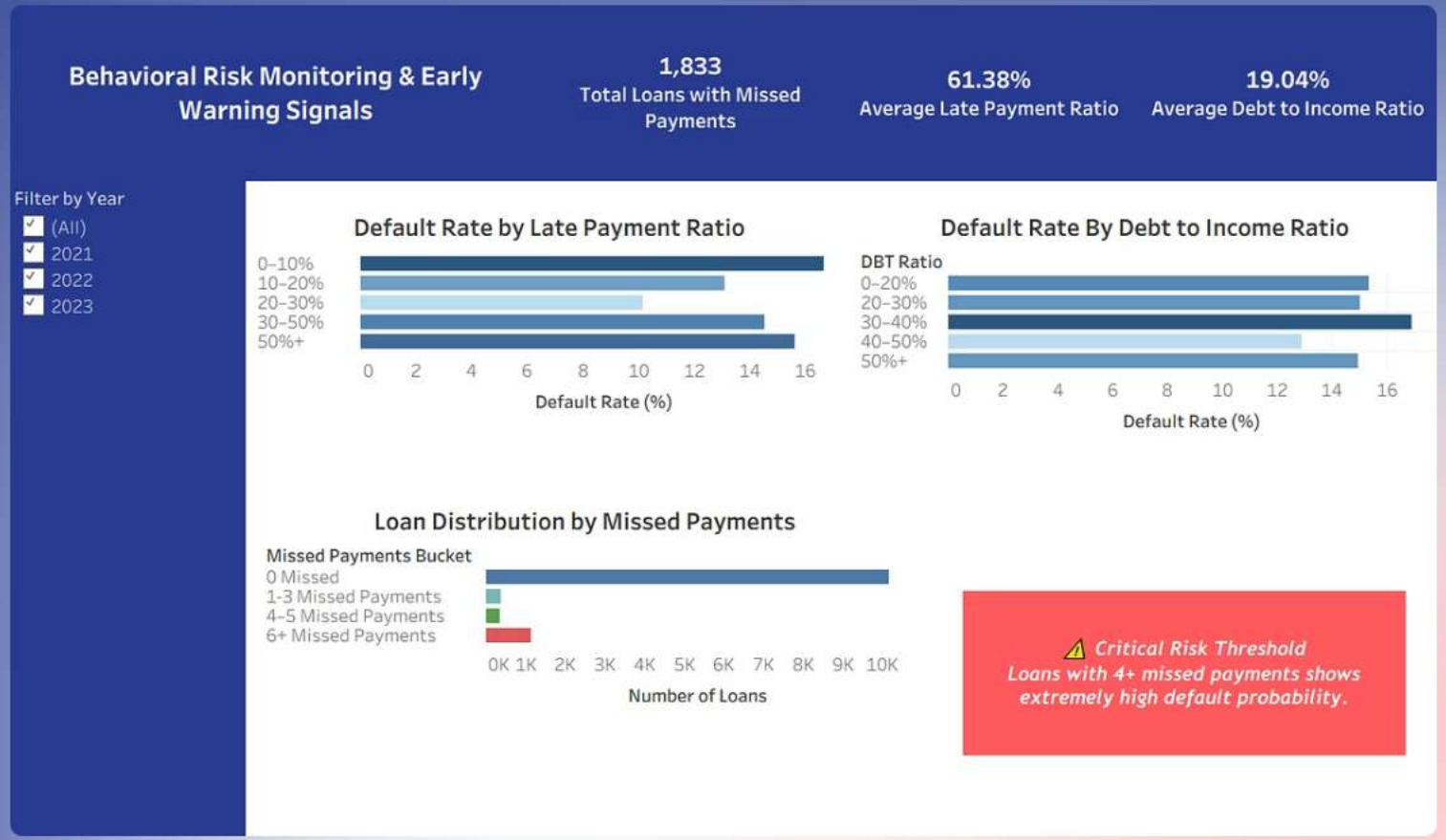
Dashboard 3: Behavioral Risk Monitoring

Purpose

Identify repayment behavior patterns that lead to loan default.

Key Metrics

- Loans with Missed Payments
- Late Payment Ratio
- Debt-to-Income Ratio



Dashboard 4: Financial Impact Analysis

₹186M

Principal Loss

Total defaulted amount

15.28%

Default Rate

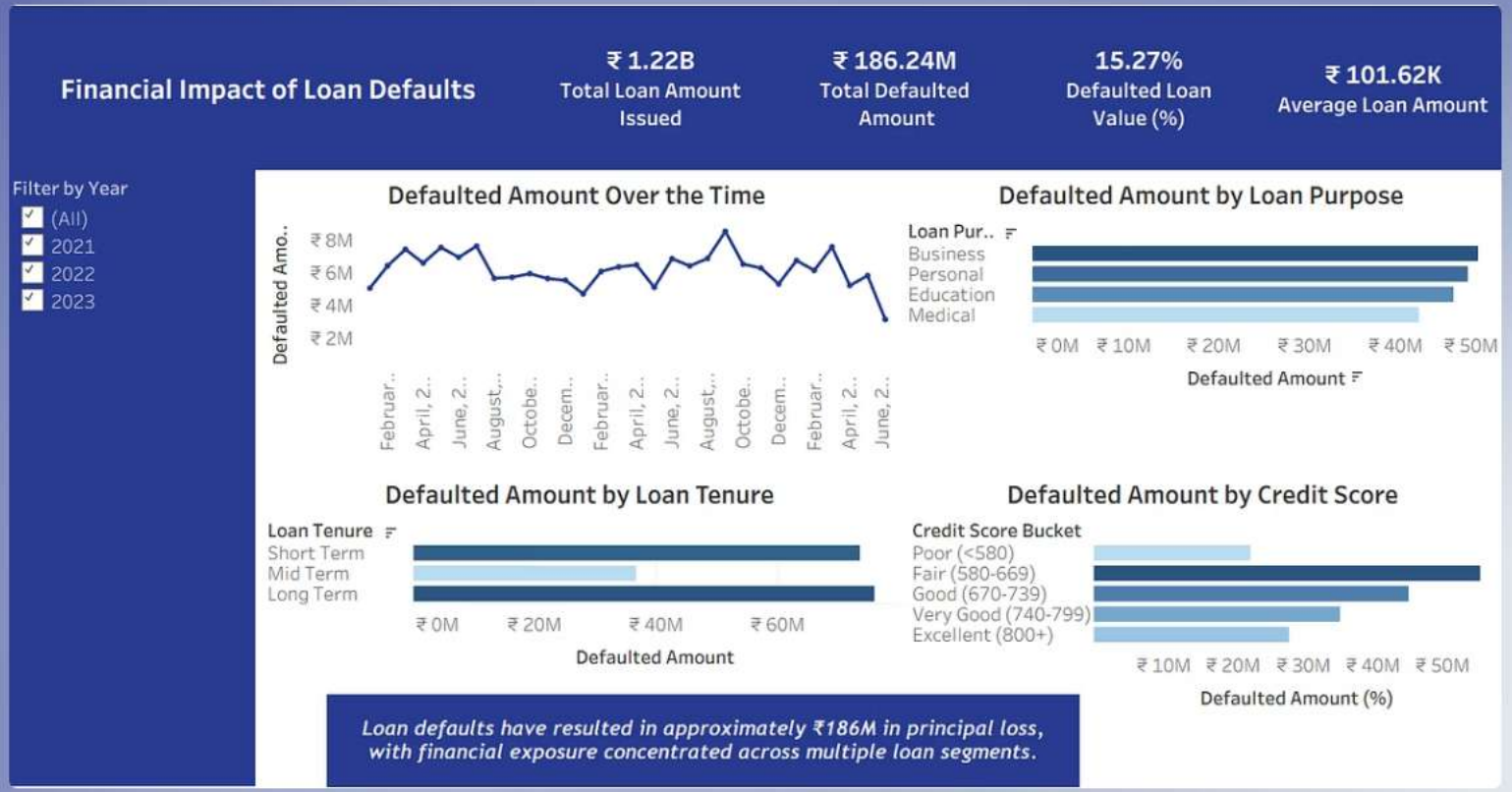
Current portfolio risk

Purpose

Measure financial loss caused by loan defaults.

Key Metrics

- Total Loan Amount Issued
- Total Defaulted Amount
- Percentage of Loan Value Lost



Key Insights & Recommendations

Key Insights

- Repayment behavior is the strongest driver of loan defaults
- Loan and customer characteristics show limited predictive power
- Default risk increases significantly after repeated missed payments
- Business loans show slightly higher risk than other loan types

Key Recommendations

- Implement early warning alerts for missed payments
- Use automated reminders to reduce missed installments
- Monitor repayment behavior continuously
- Develop behavioral risk scoring models

Conclusion & Thank You

This project demonstrates how data analysis can be used to identify key drivers behind loan defaults and support risk management decisions. The findings highlight the importance of monitoring repayment behavior and implementing early intervention strategies. By applying behavior-based risk management, financial institutions can reduce default rates, protect revenue, and improve overall portfolio performance.

> Thank You

Explore the full analysis on

GitHub

Faraz-ahmed-shaikh - Overview

IT Student | Data Analytics | Front End Development | Passionate About Creating Meaningful Insights from Raw Data | Eager to Learn and Grow in Tech - Faraz-ahmed-shaikh